ADITYA UNIVERSITY

(3)

a) Apriori Algorithm

Apriori algorithm is an algorithm used to find the frequent itemsets from the given data set of items.

*) It is a well known algorithm which is used to find the frequent itemset of the given dataset.

*) For the given set of data, we can find the frequency bought itemset of the dataset.

*) For suppose, In the field of market-basket analysis. There is a supermarket regularly the customers bought a set of items in that store.

*) Now, By using this apriori algorithm we can find the frequently bought itemset easily and we can improve the business smoothly on the basis of apriori algorithm.

*) Here we use some key keywords like support and confidence.

*) Support is the no.of frequency bought item divides the total no.of dataset.

*) By using the above formulae we can easily find the support of the given dataset.

$$\text{Support} = \frac{\text{no.of frequency of itemset (A∪B)}}{\text{Total no.of datasets}}$$

ADITYA UNIVERSITY

Example: let us consider the dataset as

I_1 = {Bread, Butter}

I_2 = {Butter, Diapers}

I_3 = {Bread, Butter}

I_4 = {Bread, Diapers}

I_5 = {Butter, Diapers}

Here, let us take individually the frequency bought of 1 itemsets initially. i.e;

Bread = 3

Butter = 4

Diapers = 3

Hence, the no.of 'butter' frequency is higher in 1 itemset.

Now, let us consider 2 itemsets

{Bread, Butter} = 2

{Butter, Diapers} = 2

{Bread, Diapers} = 1

Here both the {Bread, Butter} and {Butter, Diapers} itemsets are having highest no.of frequency sets.

*) Like this we can find the frequency bought itemset easily through the Apriori Algorithm.

x) Apriori Algorithm makes its importance in frequent itemset mining by finding the frequent itemset of the dataset from the analysis.

(6)

a) Techniques to improve classification Accuracy.

The techniques that are used to improve the functionality of the classification accuracy are

i) Feature Engineering:-

By using the concept of feature engineering the efficiency (or) performance of the classification accuracy will be improved.

ii) Data preprocessing:- There are some processes in data preprocessing that are data cleaning, data transformation, data integration... etc by this we can improve the classification accuracy.

iii) Model Selection:- By selecting various or appropriate model for the classification, we can improve the accuracy of the classification.

iv) Hyper Parameter tuning:-

By selecting the parameters that are to be used by the classification can increase the accuracy of the classification.

v) Cross validation:-

The technique of cross-validation is to ~~chee~~ cross-check the data elements so that we can improve the ~~ac~~ classification accuracy

(10)

b) Clustering Based methods in outlier detection

The clustering Based methods in outliers detection are helps to identifies the points that are different point that is different from the rest of the given data.

It works on the noise points, Boundary points and the outliers. the points that are different from the other points are called the noise points and the points which are away from the other points are called the boundary points and clustering based methods in outliers detection works on the outliers.

The methods (clustering) used in outlier detection are i) db-scan

ii) k-means

iii) k-medoids

iv) Hierarchial

These are the methods that are used in our outlier detection.

i) db-scan:

The density based clustering method which is used to works on the density of the given points & clusters the given points on the basis of the density of the points.

ii) K-means:

K-means clustering algorithm which is used to cluster the given dataset on the base of finding the centroids by the euclidean distance algorithm (or) any other distance formulas.

iii) K-medoids:

K-medoids algorithm is also an algorithm which is used in the outlier detection on behalf of clustering the points in the given dataset.

iv) Hierarchical:

It is also used in outlier detection of clustering based techniques to cluster the given points and it has some types like Agglomerative ..etc.

These are the methods that are used as an clustering based methods in outlier detection.

The steps to detect the outliers by the clustering are

- First we need to perform the clustering.
- After performing clustering, we need to identify the outliers.
- After identifying the outliers, we need to mark those outliers.

ADITYA UNIVERSITY

(8)

a) Data cleaning:-

Data cleaning is one type of data preprocessing technique that are used to clean the duplicate data (or) to keep up the space of unwanted data etc.

The techniques that are used in data cleaning are to

*) handles the missing data
*) Dealings with Outliers
*) Handling with Duplicate data
*) Standardizing the data
*) Fore up the inconsistant data
*) data passing.

* Example:-

S.no	name	marks
1.	A	50
2.	B	80
3.	C	30, 70
4.	D	-

In the above example, there are duplicate values & no values. By using this data cleaning techniques it handle the duplicate data & missing data.

Data Integration:-

Data integration is one type of data preprocessing technique which is used to integrate (or) merge the data to standardize the

ADITYA UNIVERSITY

data. The techniques that are used ~~to~~ in data integration are

*) Standardizing the data by using mean, median, mode.

*) we can take the average marks in the missing field.

*) By using this technique we can stabilize the data.

Ex:

S.no.	name	marks
1.	A	50
2.	B	30
3.	C	30
4.	D	

By using the data integration we can take (or) fill the empty space by using the mean of all the observations of the above students (or) among the students.

These are techniques that are used in data preprocessing for that we can clean the data and also we can integrate the data by the techniques of data cleaning and data integration.

vi) Logistics: In every field, we are facing the large datasets issues by using the functionalities we can easily rise up through the data mining.

vii) e-commerce:

In e-commerce, the prices may differ and we need to deal with that in any other processes.

viii) Social-media:

In social media, there are so much data is need to be handled & it makes it more efficient by knowing the techniques of data mining.

These are the main functionalities that are using Data mining inorder to functionalities these types of applications that are used to develop by using the data mining functionalities.

ADITYA UNIVERSITY

6

b) SVM:

Support vector machine is a powerful learning algorithm which is used in data mining for classification, regression and other outlier detection tasks. The support vector machine algorithm is also used in artificial intelligence on making the machines the SVM is used to solve the problems on their own on the own.

*) Advantages:

*) Effective in high dimensional
*) Efficient in memory management
*) Versatile in nature.

*) Limitations:

*) It is computationally expensive
*) It is very hard to interprete
*) It is very sensitive to noise.

K-NN:

It is the algorithm which is used to find the neighbouring algorithm used in data mining for the outlier detection using clustering the data into two (or) more parts by the process of clustering the data into itemsets that are came from the neighbouring.

Advantages:

*) It is very easy to manage

*) Least cost

*) We can easily understand.

Limitations:

*) We can't use this algorithm in a proper way.

*) Some modern techniques occupied.

*) Least effective.

There are the differences between the KNN algorithm and the support vector machine which are used to their applications.

ADITYA UNIVERSITY

(10)

a)

The difference between the global outliers and the local outliers are the global outliers which are used in anyway or anywhere in the dataset. And the local outliers which have some limitations to use wherever which is should be needed.

The global outliers are the outliers that are used globally that means we can use these global outliers wherever we can use in the outliers.

example for the global outliers is suppose there is an outlier then the outlier functionality should be used anywhere in the dataset.

The local outliers are the outliers that are used in a limited restricted area that means we can use these local outliers where ever it should be accessible.

Example for the local outliers are suppose there is an outlier then the outlier functionality should be used in a certain range in the given dataset.

Thus, the small groups are considered as samples then the functions are used in opt (or) to perform various actions on the small groups on basis of clustering to make an itemset.

These are the importance of BIRCH algorithm in clustering large datasets on basis of clustering the data into small groups. by using the BIRCH algorithm.

b)

Internal - cluster validation measures

The measures that are used to cluster the given large dataset to the small group clusters to make (or) perform several operations in order to get better results the internal - cluster validation is used. Because of by using the internal - cluster validation we can easily cluster the large datasets by internally to cluster those from the large datasets. And the measures that are used to cluster the data internally is known as the measures of internal - cluster

ADITYA UNIVERSITY

Example for the internal cluster validation measures are to cluster the large dataset, we perform the clustering technique begin with internally to make it faster.

The measures that are used to cluster the given large dataset to the small clusters to make them perform several operations in order to get the better results. The external validation is used because of by using the external cluster validation. we can easily cluster the given large dataset by externally to cluster those from the large datasets. And the measures that are used to cluster the data externally known as the measures of external-clusters.

Example for the external cluster validation measures are to cluster the large datasets, we perform the clustering techniques begin with externally to make it faster and more efficient to work.

ADITYA UNIVERSITY

③

b) Association Rule mining:-

The association rule mining is a mining techniques that are depend upon the associations which are interlinked with each other on basis of their associations.

It is safely used in the clustering of the given data and mainly used to make the data more efficient and the association rule mining are the mining techniques that are based on the associations.

On comparing with frequent itemset mining it is little bit less expressive than the frequent itemset mining. The process of analyzing the large datasets of data to find the patterns, trends and to get those useful informations example for association rule mining is to one Employee can have no.of teammates.

Frequent itemset mining:-

mining is the process of analyzing the large datasets of data and to find the patterns, trends & to get those useful information from the given large dataset.

ADITYA UNIVERSITY

Example for frequent itemset mining is to consider list of items.

I_1 = { Bread, Butter }

I_2 = { Butter, Diapers }

I_3 = { Bread, Butter }

I_4 = { Bread, Diapers }

I_5 = { Butter, Diapers }

Here, let us consider and take individually the frequency bought of a items initially i.e.,

Bread = 3

Butter = 4

Diapers = 3

Hence the no. of butter frequency is high in 1 itemset.

let us consider 2 itemset

{ Bread, Butter } = 2

{ Butter, Diapers } = 2

{ Bread, Diapers } = 1

Here both the { Bread, Butter } & { Butter, Diapers } items are have higher frequency itemset.

On comparing with Association-Rule, the frequency itemset mining is more efficient & also it is the widely used technique in data mining.